Tanmay Patil

Data Scientist

tanmayp162004@gmail.com · +91 8080380670 · Portfolio Website · Tanmay Patil · 1Tanmay6

Data Scientist building production **ML & statistical forecasting systems** (**predictive modeling, KPI automation, GAN-based synthetic data**) across supply chain, finance & insurance for enterprise clients.

Experience

Data Scientist *Bangalore, KA*
Genpact *Jun 2025 – Present*

- Built a **hybrid rule-based + ML replenishment system** (lead time, demand variability, safety stock, EOQ/RLQ) for a **top FMCG brand** – automated ∼**80%** of routine planning decisions with **100% auditability** via HITL exception handling.
- Built an **automated demand-forecasting selection engine**, evaluating **RMSE, Bias & MAPE** across forecast streams to pick the best statistical baseline per SKU – cut manual planner workload ∼**40%**.
- Architected a **financial reconciliation platform** matching bank statements, invoices & POs for a **global equipment manufacturer** – automated ∼**90%** of reconciliations, cutting manual review load ∼**90%**.
- Built a **supply-chain command center** for a **leading grocery retailer**, centralizing logistics, inventory & revenue KPIs into one interface with automated cross-team task routing.
- Built a real-time **logistics-disruption model** processing weather, news & vessel-damage alerts – ∼**85% domain accuracy**, ∼**12s** decision latency, protecting delivery SLAs.

Data Science Intern *Bangalore, KA*
Genpact *Jan 2025 – Jun 2025*

- Built a hybrid **OCR + VLM** extraction pipeline for insurance forms with a **HITL correction workflow** – cut manual effort ∼**60%** while preserving output quality.
- Built a **parallelized microservices document pipeline** (extraction, KV mapping, JSON conversion), cutting processing time **8 min → 1 min** (∼87.5%) at **78% extraction accuracy**.

Data Science Intern *Remote*
Vanquisher Software Services *Jun 2024 – Nov 2024*

- Improved a **semi-structured data-retrieval pipeline** with dynamic ranking & filtering, lifting **Precision@k by 28%** across heterogeneous document formats.
- Cut **data-processing latency** ∼**40%** by re-architecting existing pipelines for parallel execution.

Education

NIIT University *Sept 2021 – June 2025*
B.Tech in Computer Science, Specialization in Data Science · GPA: **9.56/10.0**

Projects

Hybrid Recommendation System *GitHub*

- Built a hybrid recommender (**CNN-enhanced Neural Collaborative Filtering + TF-IDF content filtering**) to resolve cold-start & data-sparsity issues, cutting **RMSE by 22%** (0.99 → 0.77) on the MovieLens 1M dataset via an 11-point ablation study.

Multi-Generator Super-Resolution GAN (MG-SRGAN)

- Designed a two-phase, multi-generator GAN architecture to fix training instability on resource-constrained hardware – lifted **PSNR 26.3%** (16.34 → 20.64 dB) over single-generator baselines and sped up training throughput ∼**30%**.

Synthetic Data Generation Research

- Benchmarked **Autoencoders vs. Transformers** for tabular and time-series synthetic data, proving autoencoder superiority for tabular synthesis; built a compliance framework guaranteeing **zero PII leakage** via differential-privacy checks.

Technical Skills

Core ML & Statistics: Python, SQL, Pandas, NumPy, Scikit-Learn, PyTorch, Statistical Modeling, Time-Series Forecasting (RMSE/MAPE/Bias), Feature Engineering, GANs (CTGAN/CopulaGAN)
Cloud, MLOps & BI: AWS (SageMaker, Bedrock, S3), Azure ML, Tableau, Power BI, Excel, KPI Dashboarding, Microservices Architecture